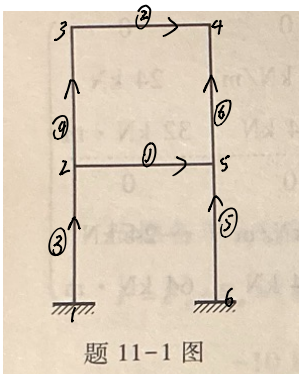
11-1

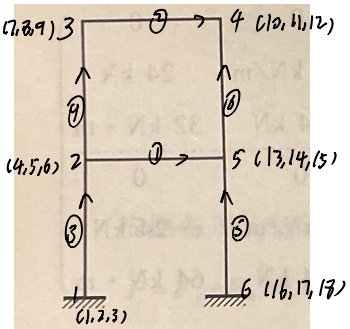
11-1　对图示刚架的单元和结点进行编号，并采用约束条件先处理法和后处理法进行位移编号。

①单元、结点编号



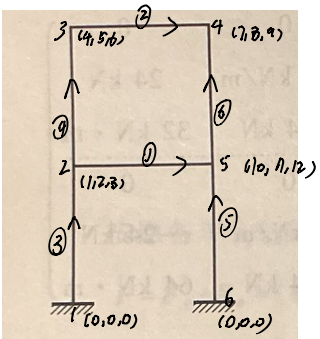
题 11-1 图

② 1° 后处理法



题 11-1 图

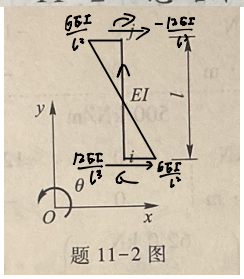
2° 先处理法



题 11-1 图

11-2

11-2　忽略轴向变形，试写出图示竖直杆单元在整体坐标系下单元刚度矩阵的第一列元素。



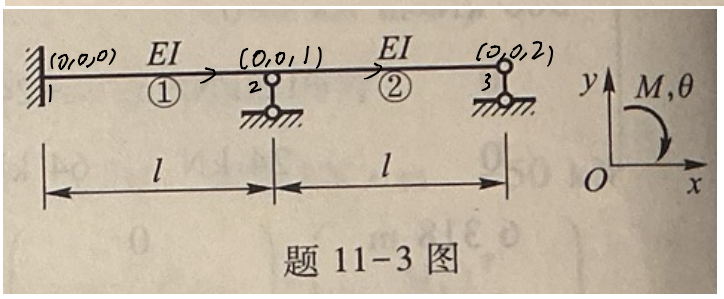
题 11-2 图

解：$\begin{bmatrix} F_{Si} \\ m_i \\ F_{Sj} \\ M_j \end{bmatrix} = \begin{bmatrix} k_{11} & & & \\ k_{21} & & & \\ k_{31} & & & \\ k_{41} & & & \end{bmatrix} \begin{bmatrix} \delta_{xi} \\ \varphi_i \\ \delta_{xj} \\ \varphi_j \end{bmatrix}$

$k_{11}=\frac{12EI}{l^3}$，$k_{21}=\frac{-6EI}{l^2}$，$k_{31}=\frac{-12EI}{l^3}$，$k_{41}=\frac{-6EI}{l^2}$

11-3

11-3　试求图示连续梁的结构刚度矩阵 $K$，已知杆件 $EI$ 为常数。



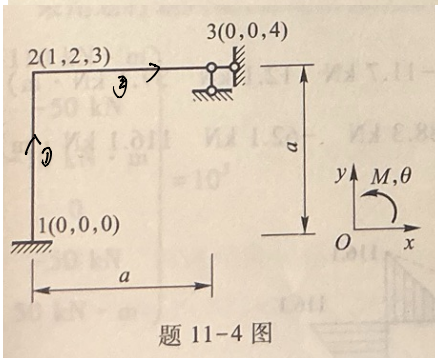
题 11-3 图

$k^{①}=\begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & k_{66} \end{bmatrix}$ 行/列：0 0 0 0 0 1

$k^{②}=\begin{bmatrix} & & & & & \\ & & k_{33} & & k_{36} & \\ & & & & & \\ & & & & & \\ & & k_{63} & & k_{66} & \end{bmatrix}$ 行/列：0 0 1 0 0 2

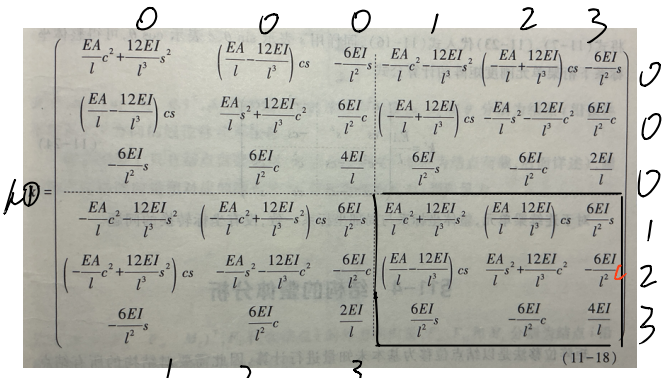
$k=k^{①}+k^{②}=\begin{bmatrix} k_{66}^{①}+k_{33}^{②} & k_{36}^{②} \\ k_{63}^{②} & k_{66}^{②} \end{bmatrix}=\begin{bmatrix} \frac{3EI}{l} & \frac{2EI}{l} \\ \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}$

11-4 11-4 试求图示刚架的结构刚度矩阵 $K$。已知 $EI=1\times10^6\ \text{kN}\cdot\text{m}$，$EA=1\times10^5\ \text{kN}$，$a=6\ \text{m}$。注：本题教材答案有错.



题 11-4 图

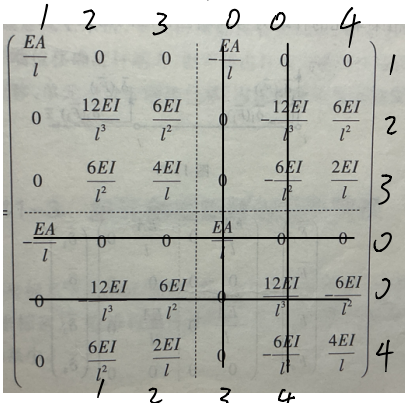
解：①结构，$\theta=90°$ 时 $\cos90°=0$，$\sin90°=1$




则

$$k^{①}=\begin{bmatrix} \frac{12EI}{l^3} & 0 & \frac{6EI}{l^2} \\ 0 & \frac{EA}{l} & 0 \\ \frac{6EI}{l^2} & 0 & \frac{4EI}{l} \end{bmatrix}\begin{matrix}1\\2\\3\end{matrix}$$

②结构 $\theta=0°$，则 $\bar{k}^{②}=k^{②}$



即

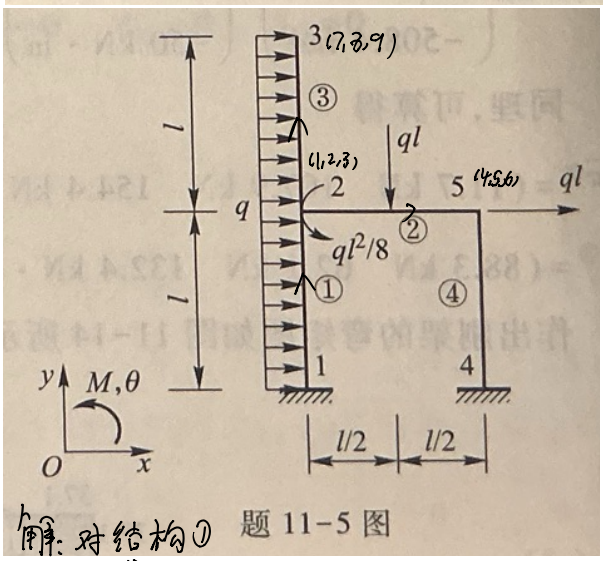
$$k^{②}=\begin{bmatrix} \frac{EA}{l} & 0 & 0 & 0 \\ 0 & \frac{12EI}{l^3} & \frac{6EI}{l^2} & \frac{6EI}{l^2} \\ 0 & \frac{6EI}{l^2} & \frac{4EI}{l} & \frac{2EI}{l} \\ 0 & \frac{6EI}{l^2} & \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}\begin{matrix}1\\2\\3\\4\end{matrix}$$

$$k=k^{①}+k^{②}=\begin{bmatrix} \frac{12EI}{l^3}+\frac{EA}{l} & 0+0 & \frac{6EI}{l^2}+0 & 0 \\ 0+0 & \frac{EA}{l}+\frac{12EI}{l^3} & 0+\frac{6EI}{l^2} & \frac{6EI}{l^2} \\ \frac{6EI}{l^2}+0 & 0+\frac{6EI}{l^2} & \frac{4EI}{l}+\frac{4EI}{l} & \frac{2EI}{l} \\ 0 & \frac{6EI}{l^2} & \frac{2EI}{l} & \frac{4EI}{l} \end{bmatrix}=\begin{bmatrix} 0.72 & 0 & 1.67 & 0 \\ 0 & 2.72 & 1.67 & 1.67 \\ 1.67 & 1.67 & 13.33 & 3.33 \\ 0 & 1.67 & 3.33 & 6.67 \end{bmatrix}$$

$\frac{EA}{l}=\frac{1\times10^8\text{N}}{6}=2.17\times10^7\ \text{N/m}$，$\frac{6EI}{l^2}=\frac{6\times10^9\text{N}\cdot\text{m}}{6^2\text{m}^2}=1.67\times10^8\ \text{N/m}$

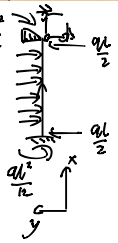
$\frac{12EI}{l^3}=\frac{12\times10^9\text{N}\cdot\text{m}}{6^3\text{m}^3}=0.55\times10^8\ \text{Pa}$，$\frac{2EI}{l}=\frac{2\times10^9\text{N}\cdot\text{m}}{6\text{m}}=3.33\times10^8\text{N}$

11-5 试求图示结构结点2的综合荷载列阵 $\boldsymbol{F}_2$。




题 11-5 图

解：对结构①

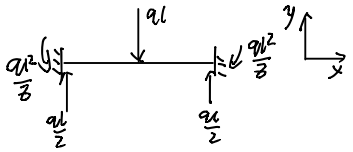


$$\bar{F}_F^{①}=(0\quad \frac{ql}{2}\quad \frac{ql^2}{12}\vdots 0\quad \frac{ql}{2}\quad -\frac{ql^2}{12})^T \qquad T^{①T}=\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&\vdots&\\0&0&1&\vdots&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}$$

$\theta^{①}=90°,\ \cos\theta=0,\ \sin\theta=1$

$$F_E^{①}=-T^{①T}\bar{F}_F^{①}=-\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&\vdots&\\0&0&1&\vdots&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}\begin{bmatrix}0\\\frac{ql}{2}\\\frac{ql^2}{12}\\0\\\frac{ql}{2}\\-\frac{ql^2}{12}\end{bmatrix}=\begin{bmatrix}\frac{ql}{2}&0&-\frac{ql^2}{12};&\frac{ql}{2}&0&\frac{ql^2}{12}\end{bmatrix}^T$$

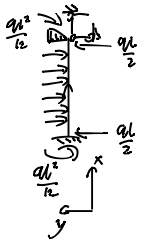
对结构②



$$\bar{F}_F^{②}=[0\quad \frac{ql}{2}\quad \frac{ql^2}{8};\ 0\quad \frac{ql}{2}\quad -\frac{ql^2}{8}]$$

$$F_E^{②}=-T^{②T}\bar{F}_F^{②}\ \text{且}\ \theta=0,\ \sin\theta=0,\ \cos\theta=1\ \text{得}\ F_E^{②}=[0\quad -\frac{ql}{2}\quad -\frac{ql^2}{8};\ 0\quad -\frac{ql}{2}\quad \frac{ql^2}{8}]^T$$

对结构③



$$\bar{F}_F^{③}=(0\quad \frac{ql}{2}\quad \frac{ql^2}{12}\vdots 0\quad \frac{ql}{2}\quad -\frac{ql^2}{12})^T \qquad T^{③T}=\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&\vdots&\\0&0&1&\vdots&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}$$

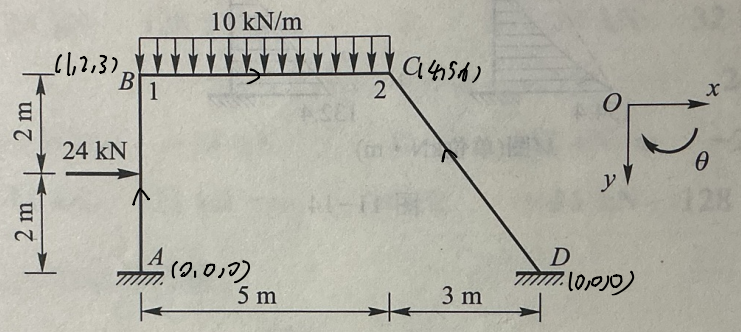
$\theta^{③}=90°,\ \cos\theta=0,\ \sin\theta=1$

$$F_E^{③}=-T^{③T}\bar{F}_F^{③}=-\begin{bmatrix}0&-1&0&\vdots&[0]\\1&0&0&\vdots&\\0&0&1&\vdots&\\\cdots&\cdots&\cdots&\cdots&\cdots\\[0]&&&\vdots&\begin{matrix}0&-1&0\\1&0&0\\0&0&1\end{matrix}\end{bmatrix}\begin{bmatrix}0\\\frac{ql}{2}\\\frac{ql^2}{12}\\0\\\frac{ql}{2}\\-\frac{ql^2}{12}\end{bmatrix}=\begin{bmatrix}\frac{ql}{2}&0&-\frac{ql^2}{12};&\frac{ql}{2}&0&\frac{ql^2}{12}\end{bmatrix}^T$$

$$F_{E2}=\begin{bmatrix}-\frac{ql}{2}\\0\\\frac{ql^2}{12}\end{bmatrix}+\begin{bmatrix}0\\-\frac{ql}{2}\\-\frac{ql^2}{8}\end{bmatrix}+\begin{bmatrix}\frac{ql}{2}\\0\\-\frac{ql^2}{12}\end{bmatrix}=\begin{bmatrix}ql&-\frac{ql}{2}&-\frac{ql^2}{8}\end{bmatrix}^T$$

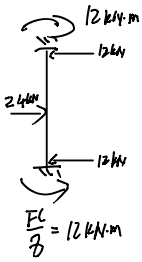
$$F_2=F_{E2}+F_p=\begin{bmatrix}ql\\\frac{ql}{2}\\-\frac{ql^2}{8}\end{bmatrix}+\begin{bmatrix}0\\0\\\frac{ql^2}{8}\end{bmatrix}=\begin{bmatrix}ql\\-\frac{ql}{2}\\0\end{bmatrix}$$

11-6 试求图示结构的综合荷载列阵。



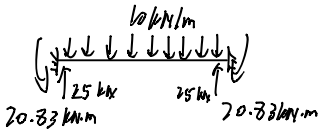

题 11-6 图

解：结构①




$$F_F^{①}=\begin{bmatrix}-12\text{kN}\\0\\-12\text{ kN·m}\\-12\text{kN}\\0\\12\text{ kN·m}\end{bmatrix}\begin{matrix}0\\0\\0\\1\\2\\3\end{matrix}$$

结构②：

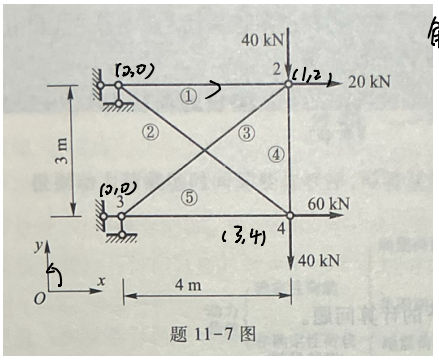



$$F_{F②}=\begin{bmatrix}0\\-25\text{ kN}\\-20.83\text{kN·m}\\0\\-25\text{kN}\\20.83\text{ kN·m}\end{bmatrix}\begin{matrix}1\\2\\3\\4\\5\\6\end{matrix}$$

$$F_E=-\bar{S}F_F=\begin{bmatrix}12\\25\\8.33\\0\\25\\-20.83\end{bmatrix}$$

$$F=F_E+F_P=\begin{bmatrix}12\\25\\8.33\\0\\25\\-20.83\end{bmatrix}$$

11-7 图示桁架的结点位移列阵 $\Delta=\frac{1}{EA}(342.322\quad -1\quad 139.555\quad -137.680\quad 1\quad 167.111)^{\mathrm{T}}$，已知各杆 $EA$ 为常数，试求单元①的内力。

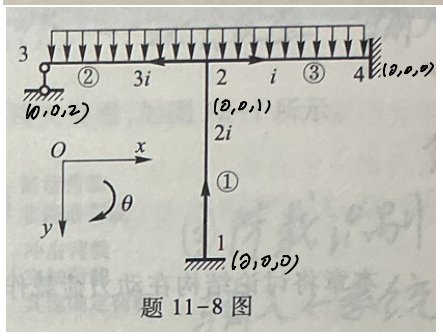



题 11-7 图

解：

$$F^{①}=\begin{bmatrix}\bar{F}_{1x}^{①}\\\bar{F}_{1y}^{①}\\\bar{F}_{2x}^{①}\\\bar{F}_{2y}^{①}\end{bmatrix}=\frac{EA}{l}\begin{bmatrix}1&0&-1&0\\0&0&0&0\\-1&0&1&0\\0&0&0&0\end{bmatrix}\cdot\frac{1}{EA}\begin{bmatrix}0\\0\\342.322\\-1139.555\end{bmatrix}=\begin{bmatrix}-85.58\\0\\85.58\\0\end{bmatrix}$$

11-8 已知图示刚架结点的转角为 $\theta_2=-\frac{1}{i}$（逆时针），$\theta_3=\frac{4}{i}$（顺时针），各杆长度 $l=6\ \mathrm{m}$，$q=14\ \mathrm{kN/m}$，试求单元①和②的杆端力



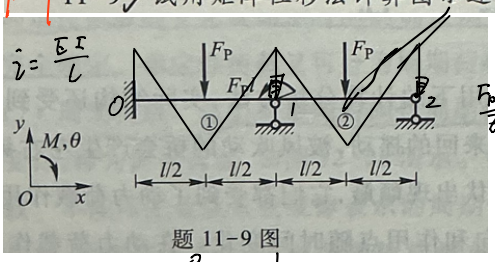
题 11-8 图

解：忽略轴向变形

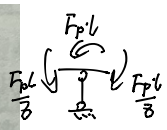
$$\bar{F}^{①}=\bar{k}^{①}\cdot\bar{\delta}^{①}=\begin{bmatrix}8i & 4i\\ 4i & 8i\end{bmatrix}\cdot\begin{bmatrix}0\\ -\frac{1}{i}\end{bmatrix}=\begin{bmatrix}-4\\ -8\end{bmatrix}$$

$$\bar{F}^{②}=\bar{k}^{②}\cdot\bar{\delta}^{②}+\bar{F}_F^{②}=\begin{bmatrix}12i & 6i\\ 6i & 12i\end{bmatrix}\cdot\begin{bmatrix}-\frac{1}{i}\\ \frac{4}{i}\end{bmatrix}+\begin{bmatrix}42\\ -42\end{bmatrix}=\begin{bmatrix}54\\ 0\end{bmatrix}$$

11-9 试用矩阵位移法计算图示连续梁的内力。已知 $EI$ 为常数。



题 11-9 图




解：$k^{①}=\begin{bmatrix}4i & 2i\\ 2i & 4i\end{bmatrix}\begin{matrix}0\\ 1\end{matrix}$　$k^{②}=\begin{bmatrix}4i & 2i\\ 2i & 4i\end{bmatrix}\begin{matrix}1\\ 2\end{matrix}$

$$F=k\cdot\Delta\quad\begin{bmatrix}M_2\\ M_3\end{bmatrix}=\begin{bmatrix}8i & 2i\\ 2i & 4i\end{bmatrix}\cdot\begin{bmatrix}\theta_2\\ \theta_3\end{bmatrix}$$

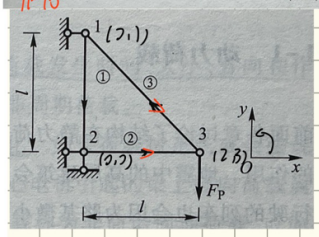
$$F=F_E+F_P=\begin{bmatrix}-F_P l\\ -\frac{F_P l}{8}\end{bmatrix}$$

$$\begin{bmatrix}\theta_1\\ \theta_2\end{bmatrix}=\begin{bmatrix}-\frac{15F_P l^2}{112EI}\\ \frac{F_P l^2}{28EI}\end{bmatrix}$$

$$F^{①}=k^{①}\delta^{①}+\bar{F}_F^{①}=\begin{bmatrix}4i & 2i\\ 2i & 4i\end{bmatrix}\cdot\begin{bmatrix}0\\ -\frac{15F_P l}{112i}\end{bmatrix}+\begin{bmatrix}-\frac{F_P l}{8}\\ \frac{F_P l}{8}\end{bmatrix}=\begin{bmatrix}-\frac{11}{28}F_P l\\ -\frac{23}{56}F_P l\end{bmatrix}$$

$$F^{②}=k^{②}\delta^{②}+\bar{F}_F^{②}=\begin{bmatrix}4i & 2i\\ 2i & 4i\end{bmatrix}\begin{bmatrix}-\frac{15}{112}\frac{F_P l^2}{6EI}\\ \frac{1}{28}\frac{F_P l^2}{6EI}\end{bmatrix}+\begin{bmatrix}-\frac{F_P l}{8}\\ \frac{F_P l}{8}\end{bmatrix}=\begin{bmatrix}-\frac{33}{56}F_P l\\ 0\end{bmatrix}$$

11-10 试用矩阵位移法计算图示桁架的内力。已知 $EA$ 为常数。




桁架 $F_P$ 均为0.

解：由图可知 $\theta_1=-90°$，$\theta_2=-45°$

$$\overline{k}^{①}=k^{①}=\frac{EA}{l}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{matrix}0\\0\\2\\3\end{matrix}$$

由 $\theta_1=-90°$ 则 $\cos\theta_1=0$ $\sin\theta_1=-1$

$$k^{①}=\frac{EA}{l}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}\begin{matrix}0\\1\\0\\0\end{matrix}$$

由 $\theta_2=-45°$ 则 $\cos\theta_2=\frac{\sqrt{2}}{2}$ $\sin\theta_2=-\frac{\sqrt{2}}{2}$

$$k^{②}=\frac{EA}{\sqrt{2}l}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{matrix}0\\1\\2\\3\end{matrix}$$

$$k=k^{①}+k^{②}+k^{③}=\frac{EA}{l}\begin{bmatrix} 1+\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & 1+\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}\begin{matrix}1\\2\\3\end{matrix}$$

$F=k\Delta$ 则

$$\begin{bmatrix} 0 \\ 0 \\ -F_P \end{bmatrix}=\frac{EA}{l}\begin{bmatrix} 1+\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ \frac{1}{2\sqrt{2}} & 1+\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} \\ -\frac{1}{2\sqrt{2}} & -\frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}\begin{bmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{bmatrix}$$

$$\begin{bmatrix} \Delta_1 \\ \Delta_2 \\ \Delta_3 \end{bmatrix}=\frac{F_P l}{EA}\begin{bmatrix} -1 \\ -1 \\ -2-2\sqrt{2} \end{bmatrix},$$

$$\overline{F}^{①}=F^{①}=k^{①}\delta^{①}=\frac{EA}{l}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ -1 \\ -2-2\sqrt{2} \end{bmatrix}\frac{F_P l}{EA}=\begin{bmatrix} F_P \\ 0 \\ -F_P \\ 0 \end{bmatrix}$$

$$F^{①}=k^{①}\delta^{①}=\frac{EA}{l}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \end{bmatrix}\frac{F_P l}{EA}=\begin{bmatrix} 0 \\ -F_P \\ 0 \\ F_P \end{bmatrix},\qquad \overline{F}^{①}=TF^{①}=\begin{bmatrix} F_P \\ 0 \\ -F_P \\ 0 \end{bmatrix}$$

$$F^{②}=k^{②}\delta^{②}=\frac{EA}{\sqrt{2}l}\begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} 0 \\ -1 \\ -1 \\ -2-2\sqrt{2} \end{bmatrix}\frac{F_P l}{EA}=\begin{bmatrix} -F_P \\ F_P \\ F_P \\ -F_P \end{bmatrix}$$

$$\overline{F}^{②}=TF^{②}=\begin{bmatrix} -\sqrt{2}F_P \\ 0 \\ \sqrt{2}F_P \\ 0 \end{bmatrix}$$

